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Marketing of ultra-processed foods on popular radio channels in Kenya

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Abstract

Background There is a dearth of information on the extent of marketing of ultra-processed foods on traditional media such as radio in low- and middle-income countries (LMICs). Consumption of ultra-processed foods is associated with nutrition-related noncommunicable diseases such as overweight/obesity, type 2 diabetes, and cardiovascular diseases. This study aimed to examine the marketing of ultra-processed foods on the most widely accessed radio stations across three counties in Kenya.

Methods This was a cross-sectional quantitative study conducted in three counties: Nairobi, Mombasa, and Baringo counties in the period between December 2021 to February 2022. We purposively selected 5 radio stations based on their popularity in different counties according to the Communications Authority of Kenya ratings. Using stratified sampling, we selected 8 recording days: 4 weekdays and 4 weekend days for three months. The recorded data were coded using a structured questionnaire. The key variables included the food and non-food and beverage products advertised on the radio stations, categorization of the food, non-food beverage products, time slots of the advertisements, promotional strategies, and premium offers.

Results Of the 1499 advertisements on the radio, 15.7% ($n = 235$) were food and beverage products. The most advertised food categories were sugar-sweetened beverages (44.7%) and alcoholic drinks (23.4%). Ultra-processed foods (UPFs) accounted for 58.3% of the non-alcoholic food and beverage product advertisements on the radio. There was a significantly higher level of UPFs during the weekdays (58.3%) compared to weekend days (39.0%) ($p < 0.001$), school holiday seasons (73.4%) compared to non-school holiday seasons (46.5%) ($p < 0.001$), and in urban areas (70.1%) compared to rural areas (34.5%) ($p < 0.001$).

Conclusions We observed a high level of UPF marketing across both rural and urban settings, with higher proportions recorded in urban areas and during school holiday periods. To support healthier food environments, there is a need for regulatory action targeting radio advertising of unhealthy foods. This may include restrictions during peak child listening hours, regulation of persuasive content, and clear identification of sponsored food promotions.

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Keywords Media, Radio, Traditional media, Marketing, Ultra-processed foods, Unhealthy foods, Processed foods, Policies, Food environments, Listenership, Dietary behaviors, Exposure, Promotion, Promotional strategies, Premium offers

Background

The rising burden of NCDs presents a significant public health challenge in Kenya, with cardiovascular diseases contributing the largest proportion of NCD-related deaths (59%), followed by respiratory illnesses (14%) and diabetes (12%) [1, 2]. According to the 2022 Kenya Demographic and Health Survey (KDHS), 27% of adults are classified as overweight or obese, with a higher prevalence among women (38.5%) compared to men (17.5%) [3]. Among adolescent girls aged 15–19 years, 18% are underweight, while 13% present with overweight or obesity, comprising 11% overweight and 2% [3]. The marketing of unhealthy foods has emerged as a significant public health concern globally, particularly in low- and middle-income countries (LMICs) where rapid urbanization and economic development have led to shifts in dietary patterns. The World Health Organization (WHO) has identified unhealthy diets as a major risk factor for nutrition-related non-communicable diseases (NCDs), which are responsible for a substantial burden of morbidity and mortality worldwide [4, 5].

Mass media campaigns on various media platforms are associated with behavior change and have been used in various settings in LMICs for health messaging [6–8]. However, audiovisual recording broadcasting often also includes advertisements for unhealthy foods, which can influence consumer behavior and contribute to poor dietary choices [8]. In Kenya, as in many other LMICs, the use of digital media, information and communication technology (ICT), internet connectivity, radio, and television (TV) negatively influences dietary choices and is associated with overnutrition, overweight/obesity, showing significant public health implications [9]. Traditional media, including radio, remains a powerful tool for communication in Kenya, particularly in rural areas where access to the internet and digital platforms is limited [10]. Currently, 75% of Kenyans still listen to the radio according to the Kenya National Media Status Report 2024, and the listenership is higher among males compared to females, more in rural areas compared to urban areas, and highest among older ages [10]. Additionally, communications authority surveys conducted in 2023 elucidated that radio listenership was estimated to be 71%, 72%, 76%, and 83% for age groups 15–17 years, 18–24 years, 25–34 years, and 35–44 years, respectively [11]. Radio broadcasts in Kenya are therefore reaching a wide audience, including children and adolescents, making them an effective medium for disseminating information about food products.

Globally, the marketing of foods has been linked to increased consumption, choice, and preference of food products, particularly among children and adolescents [12]. Studies have shown that exposure to food advertising can lead to increased preferences for unhealthy food options, ultimately contributing to the obesity epidemic [13]. Unhealthy foods and beverages are defined as products that are high in nutrients of concern, including added sugars, sodium (salt), and saturated or trans fats, and are typically energy-dense with minimal nutritional value [14]. In LMICs, the situation is exacerbated by the lack of stringent regulations governing food marketing practices, which often allows for the aggressive promotion of unhealthy products [15, 16]. This is particularly relevant in the context of Kenya, where the regulatory framework surrounding food marketing is still developing, and there is a pressing need for research to understand the extent and impact of unhealthy food marketing on public health. The impact of such marketing practices is particularly concerning given the rising prevalence of obesity and related health issues in Kenya.

Despite the growing body of literature on food marketing and its health implications, there remains a significant research gap in the context of Kenya and other LMICs. Emerging evidence from low- and middle-income countries highlights the growing influence of food marketing on adolescent dietary behaviours, with studies documenting the use of price promotions, emotional appeal, and teen-oriented digital strategies to promote fast food and shape food preferences and consumption patterns across various media platforms [17, 18]. Most existing studies have focused on high-income countries, leaving a dearth of information on how traditional media marketing influences dietary behaviors in LMICs. Current studies in Kenya and comparable countries in Africa, such as Ghana and Uganda, have elucidated high levels of marketing and provision of unhealthy foods in various settings, including around schools and in modern retail outlets [19–23]. Further studies were conducted on TV media in Nigeria and South Africa, where there was a high level of unhealthy food and beverage advertising, especially for sugar-sweetened beverages such as fruit juice and carbonated drinks, fatty foods, and desserts [24, 25]. Despite the exploration of marketing in various food environments in LMICs, traditional media, particularly radio, have received less attention.

The current study aims to address this gap by examining the extent of marketing of UPFs on the most common radio channels in three counties in Kenya. The findings

will contribute to a better understanding of the marketing landscape for unhealthy foods in LMICs and inform public health strategies aimed at mitigating the adverse effects of unhealthy food marketing.

Methods

Study design

This was a cross-sectional quantitative study conducted in three counties in Kenya: Nairobi, Mombasa, and Baringo, in the period between December 2021 to February 2022. We used structured questionnaires to extract quantitative radio-recorded data from the most popular radio stations in the three counties. Following the International Network for Food and Obesity, Non-communicable Diseases Monitoring and Action Support (INFORMAS) guidelines, we formulated the tools and field manuals strictly following instructions on monitoring the marketing of unhealthy foods on digital media [26].

Sampling procedures and selection of radio stations

The selection of the counties was done purposively to have a representative sample of rural (Baringo county) and urban settings (Nairobi and Mombasa counties) in Kenya. The methodology entailing the selection of the counties is outlined by a study conducted in Kenya to assess the marketing and availability of unhealthy foods around school environments and modern retail outlets [20, 21]. We utilized a purposive sampling method to select radio stations that were commonly listened to by the general population in the three counties, as stipulated by the Communication Authority of Kenya's (CAK) 2021 and the State of the Media report 2021–2022 [27, 28]. We selected Citizen Radio and Jambo FM for Nairobi county, Radio Maisha, Jambo FM, and Kaya FM for Mombasa County, and Kass FM for Baringo County, out of the 10 nationally listened to radio stations [29]. Despite Jambo FM, Radio Maisha, and Citizen Radio being popular in Nairobi county and Mombasa counties, these stations do reach national audiences as they are listed among the most popular radio stations in the country [29]. These radio stations were selected based on their popularity by the general population in the individual counties based on the CAK and media reports ratings. We used stratified random sampling to allocate recording days into the weekday and weekend day categories using INFORMAS guidelines. All days within the three-month monitoring period were listed and stratified into weekdays (Monday to Friday) and weekend days (Saturday and Sunday). From each category, four days were randomly selected, resulting in a total of eight recording days that ensured balanced representation of both weekday and weekend advertising patterns. We included the school holiday seasons and non-school holiday seasons to identify the trends of marketing food and beverage

products by school seasons [26]. School holiday periods were included in the analysis to explore potential patterns in ultra-processed food advertising that may indicate child-directed marketing, although children were not the exclusive focus of the study. We excluded national public holidays, including Christmas Day and New Year's Day, from the selected days to ensure that radio programming recorded in the three counties reflected the regular broadcast schedule.

Radio data recording, data coding, and extraction

We recruited 5 research assistants in the three counties and assigned them individual radio stations to record. They were equipped with digital radios and recorders to facilitate recording of the stations. The recording in each station was scheduled for 18 h between 6 am and 12 midnight on the selected days. The recorded data were saved and uploaded on Google Drive for download and archiving on hard disks and laptops by the project team to avoid any losses. A data extraction tool was developed in a survey format and was programmed in SurveyCTO to record essential variables needed for the study. The extraction entailed all the non-food and food and beverage advertisements in the defined time slots. The second step was identifying the individual product. The individual food and beverage items were extracted and further categorized into different food categories at the analysis stage. A daily monitoring tool was developed in the Kobo Toolbox applications to follow up on the progress of recording and challenges faced, including interruptions and other technical difficulties. The extracted data were synced daily, and the progress was monitored weekly.

Description of variables

All the advertisements, both food and beverage advertisements, and non-food and beverage advertisements were recorded, with the respective time slots and radio programs. The variables extracted included the date of recording, time slot of the advertisement, county, type of advertisements (food and beverage advertisements, and non-food and beverage advertisements), specific item advertised, food category advertised, any persuasive or promotional strategies, and premium offers in the food and beverage advertisements. The persuasive techniques included the use of child-like language and jingles in advertisements, the use of famous sports personalities, sports, and historical events, while the premium offers included free application downloads, price offers, discounts, contests, and charity. Reliability checks were conducted on selected channels in specific time slots by a pair of research assistants to assess the credibility of the coded data. The credibility was assessed by the proportion of similarity in the data extracted using the level of

agreement measure. The level of agreement was set at 80%.

Data analysis

The foods were categorized into individual food groups and later categorized based on the level of processing according to the NOVA classification [23, 30]. Alcoholic beverages were not included in the NOVA classification applied in this study, as they are not consistently defined as ultra-processed products. Many alcoholic drinks, such as beer, wine, and spirits, are produced through fermentation or distillation without the addition of sugars or additives and therefore do not meet the criteria for ultra-processed foods within the NOVA framework. The remaining food and non-alcoholic beverages were categorized into (i) unprocessed or minimally processed foods, (ii) processed culinary ingredients, (iii) processed foods, and (iv) ultra-processed foods (UPFs). The unprocessed/ minimally processed foods and processed culinary ingredients represented the healthier food options with a low level of processing and are either in their natural state or contain a low level/ recommended nutrient of concern, such as salts, fats, and sugar [28]. The processed culinary ingredients refer to substances derived from natural foods through pressing, refining, or grinding, and are typically used in home or restaurant food preparation (e.g., oils, sugar, and salt [28]). Processed foods are products made by adding salt, sugar, or other substances to unprocessed foods to preserve them or enhance their flavor (e.g., canned vegetables, cheeses, and artisanal bread [28]). The UPFs are unhealthier food options, which have undergone a series of industrializations and contain high levels of nutrients of concern, such as sugar and salts [28]. We calculated the mean rate of food and non-food advertisements across radio channels and used descriptive frequencies (%) to present the distribution of food categories by type. To assess statistical differences across independent variables such as season, county, and day of the week in NOVA food classifications, we applied the non-parametric Kruskal–Wallis test. This test was appropriate as the outcome variables, advertisement frequencies, were non-normally distributed count data, and the independent variables included ordinal or discrete

groupings, such as days of the week, counties, and seasonal categories. The Kruskal–Wallis test allowed for comparison across more than two independent groups without assuming normality or homogeneity of variances. Stata Version 18 was used for analysis, and the statistical significance level was set at $\alpha = 0.05$.

Results

General description of food and non-food advertisements

We observed 1,499 total advertisements in different radio stations within the 8 recording days. We recorded a total of 144 h per radio station for the 18-hour-a-day recording time and a total of 720 h in the five radio stations. The overall mean rate of advertisements on radio was 2.1 per hour, while the mean rate of advertisements for food and beverage products and non-food products was 0.3 and 1.8 per hour, respectively. Of all the advertisements on the radio, 15.7% ($n=235$) of these advertisements were food and beverage products.

The non-food and beverage advertisements were significantly more frequent compared to the food and beverage advertisements on various radio stations ($p < 0.0001$). Most of the non-food advertisements were recorded on Citizen Radio ($n=490$, 38.8%), followed by Jambo FM ($n=428$, 33.9%), and least on Radio Maisha ($n=50$, 4.0%) (Table 1). The advertisement of food and beverage products was observed highly in Kaya FM ($n=61$, 26.0%), followed closely by Kass FM ($n=58$, 24.7%) (Table 1). The mean rate of the overall advertisements was highest on Citizen Radio (mean rate = 1.2), followed by Jambo FM (mean rate = 1.1) (Table 1).

Overall, food and beverage advertisements on radio stations

The most commonly advertised food categories on the radio were sugar-sweetened beverages/ drinks ($n=73$, 31.1%), and alcoholic beverages ($n=55$, 23.4%) (Table 2). There was a substantial level of advertisements for bread, rice, and cereal/grain products with added fat, sugar, and salt products ($n=44$, 18.7%) and tea and coffee products with no additives/sweeteners ($n=17$, 7.2%) (Table 2). Considering the NOVA classification, the food categories that contributed to the UPFs included Sugar-sweetened

Table 1 Distribution of food and non-food advertisements by channels

	Food advertisements <i>N</i> = 235 <i>n</i> (%)	Non-food advertisements (<i>N</i> = 1264) <i>n</i> (%)	<i>P</i> -value	Mean Rate per channel hour (Non-food)	Mean rate per channel hour (Food)
Citizen Radio	35 (14.9%)	490 (38.8%)	< 0.001	3.4	0.2
Radio Maisha	30 (12.8%)	50 (4.0%)		0.3	0.2
Kass FM	58 (24.7%)	124 (9.8%)		0.9	0.4
Jambo FM	51 (21.7%)	428 (33.9%)		3.0	0.4
Kaya FM	61 (26.0%)	172 (13.6%)		1.2	0.4

**Notes: *P*-value from Kruskal–Wallis test

Table 2 Overall food categories using NOVA classification

Food Categories	Frequency (N=235)	Percentage
Unprocessed/minimally Processed Foods	44	(18.7%)
Bread, rice, noodles, and cereal/grain products without added fat, sugar, or salt.	44	(18.7%)
Processed Culinary Ingredients	29	(12.3%)
Tea and coffee products with no additives/sweeteners	17	(7.2%)
Oils high in mono- or polyunsaturated.	12	(5.1%)
Processed foods	2	(0.9%)
Bread, rice, and grain such as millet products with added fat, sugar, salts, and vitamins (fortified)	2	(0.9%)
Ultra-Processed Foods	105	(44.7%)
Sugar-sweetened drinks/beverages - include soft drinks and energy drinks	73	(31.1%)
Fast foods, not only healthier options advertised	13	(5.5%)
Recipe additions-highly processed/not in the natural state	11	(4.7%)
Tea and coffee products with sweeteners/additives	8	(3.4%)
Alcohol	55	(23.4%)

Notes: Recipe Additions- Highly processed spices with high levels of salts and additives

Table 3 Advertisements of foods by holiday seasons, day of the week, and County

Variables	Description	NOVA food categories				P-Value
		U/MPF n (%)	PCI n (%)	PR n (%)	UPF n (%)	
Day of the week	Overall	44(24.4%)	29(16.1%)	2(1.1%)	105(58.3%)	0.0001
	Mean rate per hour	0.1	0.04	-	0.15	
	Weekend	25 (32.5%)	20(26.0%)	2(2.6%)	30(39.0%)	
	Weekday	19(18.5%)	9(8.7%)	0	75(75.8%)	
Season	School holiday season	6(7.6%)	13(16.5%)	2(2.5%)	58(73.4%)	0.0001
	Non-school holiday season	38(37.6%)	16(15.8%)	0	47(46.5%)	
County	Nairobi	0	9(25.7%)	2(5.7%)	24(68.6%)	0.0001
	Mombasa	14(16.1%)	12(13.8%)	0	61(70.1%)	
	Baringo	30(51.7%)	8(13.8%)	0	20(34.5%)	

Notes: U/MPF-Unprocessed/Minimally processed foods, PCI-Processed culinary ingredients, PR-Processed foods, UPF-Ultra-processed foods. P-Values- Kruskal-Wallis tests output

drinks/ beverages - including soft drinks and energy drinks ($n = 73$, 40.6%), fast foods ($n = 13$, 7.2%), and recipe additions ($n = 11$, 6.1%). Considering all food and beverage products, UPFs accounted for 44.7% of all food and beverage products advertised on the radio. Furthermore, there was a considerable advertisement of healthier options in the unprocessed/minimally processed category ($n = 44$, 18.7%), and processed culinary ingredients category ($n = 29$, 12.3%), which were mainly composed of tea products ($n = 17$, 9.4%) and cooking oils ($n = 12$, 6.7%). Considering the UPFS and the alcohol advertisements, which are health-harming, these products accounted for 68.1% of all food and beverage products, hence showing a significant level of advertisements of unhealthy foods on traditional media. Excluding the Alcoholic beverages ($n = 35$), approximately 58.3% of the non-alcoholic food and beverage products ($n = 180$) were in the UPF category.

Food advertisements during school holidays and non-holiday seasons, county, and weekend/weekend days

There were significant differences between the day of the week and the advertisement of different food categories

($p < 0.001$), where we observed a higher level of advertisements of UPFs during the weekdays ($n = 75$, 75.8%) compared to the weekend days ($n = 30$, 39.0%) (Table 3). We observed a significantly high proportion of advertisements of UPFs during the school holiday seasons ($n = 58$, 73.4%, $p < 0.001$), and in Mombasa county ($n = 61$, 70.1%) and Nairobi county ($n = 24$, 68.6%) compared to the rural Baringo county ($n = 20$, 34.5%) ($p < 0.001$).

Trends of advertisement of foods using the NOVA classification in the 18-hr period

We observed higher advertising numbers for UPFs in most time slots compared to other food categories (Fig. 1). The advertisements for UPFs were highest between 11 a.m. and 5 p.m. and drastically dropped from 7 p.m.

Use of marketing strategies and promotional offers

The most used promotional strategies in food and beverage advertisements on radio were historical events ($n = 67$), movie titles ($n = 29$), and sports events ($n = 6$). Premium offers included gifts or collectibles ($n = 20$),

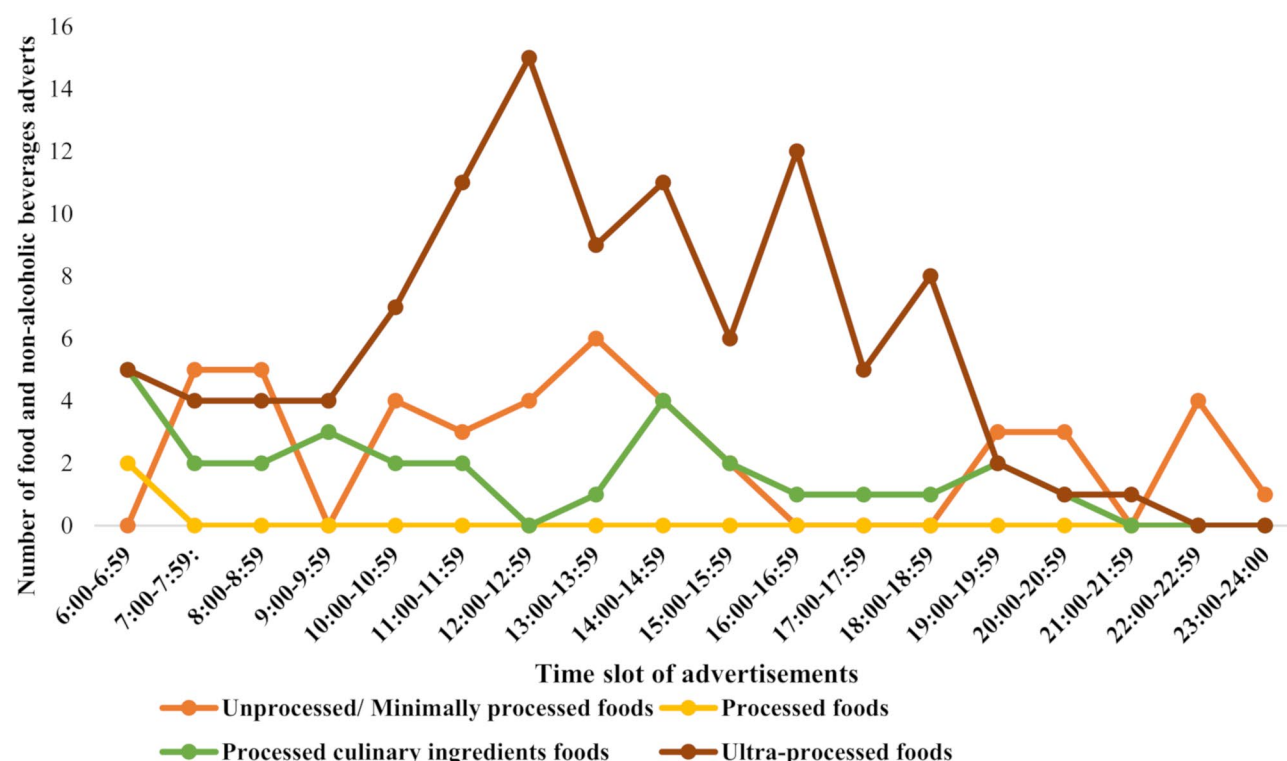


Fig. 1 Trends in the marketing of food and non-alcoholic beverages by NOVA classes

game application downloads ($n=5$), social charity ($n=5$), 20% extra ($n=2$), and contests ($n=1$).

Discussion

Our study aimed to assess the extent of advertisements for ultra-processed foods on radio stations across rural and urban counties in Kenya. The key findings indicate that there is a high level of advertisement of these foods, raising concerns about their potential impact on public health, especially considering the growing burden of nutrition-related NCDs in Kenya. We also observed a significantly high level of advertisements of UPFs during weekdays, school holiday seasons, and in urban areas.

The high frequency of ultra-processed food advertisements found in our study is consistent with trends observed in other regions globally [31]. Specifically, the increased frequency of food advertisements during school holiday periods suggests heightened exposure among children, a group that is particularly susceptible due to limited cognitive ability to critically interpret persuasive marketing content, making them more vulnerable to the influence of unhealthy food promotion. Similarly, studies in the United States and the United Kingdom have highlighted the overwhelming presence of unhealthy food advertisements, particularly those targeting children and vulnerable populations [32, 33]. Furthermore, in Australia, the advertisements of these foods in various settings were positively correlated with increased purchases and

dietary behaviors such as high intake of unhealthy drinks [34]. In LMICS, including Kenya, similar patterns have been observed where the proliferation of advertisements of UPFs was observed in substantial proportions in other settings, such as outdoor advertisements around schools and in modern retail stores [19–23]. The unified rising levels of unhealthy food advertising across diverse settings may reflect broader patterns of urbanization and economic growth, which are increasingly extending into rural areas in low- and middle-income countries. The prominence of such advertisements in various settings, including traditional media platforms such as radio, an accessible medium with broad reach, raises concerns about the influence they might have on consumer behavior, especially in rural and low-income areas where access to health information might be limited.

We observed a higher level of advertisements of UPFs in urban areas compared to rural areas in the radio recordings. This finding is consistent with previous studies that have shown disproportionate marketing and availability of unhealthy foods across different socioeconomic groups and between rural and urban areas. In India, energy-dense, unhealthy snack foods with saturated fats were prominent in rural low socioeconomic settings and urban informal settlements [35]. This contrasts with our findings, which revealed a higher intensity of UPF marketing in urban areas, likely attributable to higher radio listenership in urban settings of Kenya

compared to rural areas. In contextualized settings in Ghana and Uganda, there was a significant level of marketing of unhealthy foods around other settings, including schools and supermarkets; however, rural-urban comparisons were not performed [19, 22, 23]. In Kenya's modern retail outlets, a recent study elicited a significantly high level of availability and marketing of UPFs in coastal-urban areas compared to rural areas [21]. Moreover, an assessment of the level of marketing of foods on TV media in African countries has shown a substantially high level of marketing of UPFs, including sweetened drinks and fast foods, in peak viewing times for children and adolescents [24, 25]. This finding underscores the continued prominence of radio as a key channel for disseminating information and promoting unhealthy foods to both children and adults in the Kenyan context. In contrast, a study conducted in the same Kenyan settings around urban and rural schools showed a higher rate of advertisements of UPFs in rural Baringo county and low socioeconomic areas [20]. Additionally, these trends in urban areas are expected due to the progressive urbanization, which has necessitated a nutrition transition in LMICs, hence enhanced availability, marketing, and consumption of foods high in nutrients of concern, such as fats and sugars [36]. This is an indication that, just as observed in traditional media, specifically the radio, the same is happening in other food environment settings such as school neighborhoods and both formal and informal outlets in the country. Our findings suggest that the high presence of such advertisements in diversified settings might be contributing to unhealthy dietary patterns.

Additionally, the radio's broad reach in Kenya, including among underserved rural populations, could make these advertisements particularly influential. This is important because radio continues to be a leading source of information and entertainment in many parts of Kenya, particularly in areas where television and internet access are limited [10, 27]. The rural population's exposure to these food ads without adequate nutrition education or regulation could contribute to a greater reliance on unhealthy food choices, further exacerbating the country's nutrition transition from traditional, nutrient-dense diets to more processed, calorie-dense ones.

Furthermore, while advertisements for unhealthy foods are prevalent, there appears to be a lack of regulatory frameworks in Kenya to manage the content and timing of such advertisements. Internationally, countries like Chile and Mexico have implemented regulations to curb the marketing of unhealthy foods, particularly to children, as a strategy to reduce the impact of food marketing on diet-related diseases [37–40]. In Kenya, the absence of similar regulations allows for unrestricted advertising, which might be hindering efforts to promote healthier dietary patterns. A national policy or regulatory

framework, such as those proposed by the World Health Organization (WHO), could help mitigate the effects of such advertisements by restricting unhealthy food ads, especially during times when children and adolescents are most likely to be listening. The human rights-based approach for safeguarding the health and well-being of children can additionally enhance more stringent measures in different food environments as recommended by UNICEF [41, 42]. Another important consideration is the broader socioeconomic implications of promoting UPFs. These foods are often marketed as affordable, convenient, and accessible, which can influence consumer choices, particularly in lower-income communities. However, this emphasis on convenience and price overlooks the long-term health costs associated with poor dietary habits, which can lead to a higher burden on the healthcare system as more individuals develop diet-related diseases [43–45]. Given the increasing rates of NCDs in Kenya, the promotion of unhealthy food products could exacerbate the financial strain on both individuals and the national healthcare system, highlighting the need for a holistic approach that incorporates both regulatory measures and public health education.

Strengths, limitations, and recommendations

While our study provides valuable insight into the prevalence of unhealthy food advertisements in Kenya, it is limited by its focus on radio alone. Future studies should consider analyzing other forms of media, such as digital platforms, to provide a more comprehensive view of the landscape of unhealthy food marketing in Kenya. This study is novel and adds to the ongoing research work in different food environment settings in Kenya. As the first of its kind in Africa, we observe differences in the extent of marketing unhealthy foods in rural and urban settings in Kenya using traditional media such as radio. Our study findings suggest the need for the implementation of policies that regulate food advertising, particularly those targeting children and low-income populations, in line with global best practices. To mitigate the public health risks associated with such exposure, particularly among children and adolescents, it is recommended that Kenya adopt time-based restrictions on unhealthy food advertising during peak listening hours. Additionally, policies mandating the disclosure of sponsored food content on the radio should be considered to enhance transparency. These measures should be complemented by a comprehensive national food marketing policy and mass media nutrition education campaigns to support healthier food choices and reduce the influence of UPF marketing across both rural and urban populations. Future research should consider longitudinal assessments of ultra-processed food advertising trends and their association with dietary behaviours and nutrition-related health outcomes, such

as overweight, obesity, and type 2 diabetes, to better understand the long-term public health implications of food marketing in Kenya.

Conclusions

This study found a high frequency of ultra-processed food (UPF) advertisements on radio stations in Kenya, with significantly greater exposure during weekdays, school holiday periods, and in urban areas. These patterns are particularly concerning in light of the country's rising burden of nutrition-related non-communicable diseases (NCDs). The findings highlight the need for urgent regulatory measures to curb the marketing of unhealthy foods, especially through widely accessed platforms such as radio. A comprehensive policy response is warranted; this may include time-based restrictions during peak child listening hours, regulation of persuasive advertising techniques, mandatory disclosure of sponsored content, and the development of a national food marketing framework. Complementary public health education efforts are also needed to promote healthier dietary choices and support a more health-enabling food environment across both rural and urban settings.

Abbreviations

NCDs	Non-communicable diseases
NR	NCDs-Nutrition-related non-communicable diseases
WHO	The World Health Organization
LMICs	Low and middle-income countries
UPFs	Ultra-processed foods
ICT	Information and communication technology
APHRC	African Population and Health Research Centre
MOH	Ministry of Health
CAK	Communication Authority of Kenya's (CAK)
ERSC	Ethics and Scientific Review Committee
NACOSTI	The national commission for science, technology, and innovation

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40795-025-01164-5>.

Supplementary Material 1

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Author contributions

G.A., P.I., C.A., A.L., M.H., K.G., and S.V. were involved in the conceptualization and design of the study. C.H.K., S.V., M.N.W., and G.A. developed the analysis plan. C.H.K. analyzed the data with support from M.N.W., S.V., C.A., and G.A. M.N.W., C.H.K., V.O., and S.M. supported project implementation, management, project monitoring, and data collection. G.A., C.A., S.V., A.L., M.H., and K.G. contributed to the synthesis and interpretation of the results. C.H.K. and M.N.W. developed the first manuscript draft. G.A., S.V., C.A., A.L., K.G., and M.H. supported further input and edits on the manuscript. All the authors contributed to the finalization of the manuscript. All authors approved the final manuscript.

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Data availability

Data and other materials from this study will be available upon request by other researchers. The primary author will be the contact person for the provision of data and other supporting documents related to the work.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the AMREF-Health Ethics and Scientific Review Committee (ESRC) in Kenya (ESRC/P901/2020) and the National Commission for Science, Technology, and Innovation (NACOSTI) (NACOSTI-P/22/19104). This research was performed in accordance with the Declaration of Helsinki.

Consent to participate

No consent was obtained as no human participants were included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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